

SMART MANUFACTURING • INDUSTRIAL ENERGY & PROCESS EFFICIENCY

FactoryMind

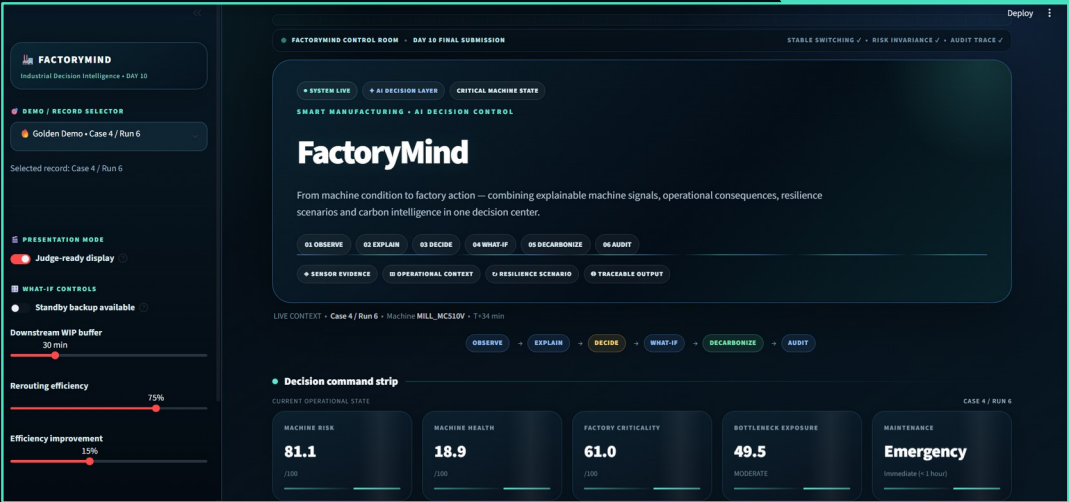
From machine risk to factory decarbonisation.

An operating decision center for Indian SMEs that connects machine condition, factory consequence, energy efficiency and carbon intelligence in one loop.

OBSERVE → EXPLAIN → DECIDE → OPTIMIZE

Built for the decision moment

Not just “is the machine risky?” — but “what should the factory do next?”



LIVE PRODUCT VIEW • GOLDEN DEMO / CASE 4 • RUN 6

TEAM FLUXFORGE

Ronit • Yogik • Rakesh • Shrikant

The factory does not need another alert — it needs a better decision.

Challenge 04 is about affordable, real-time energy visibility, predictive maintenance, process optimisation and decarbonisation without sacrificing competitiveness.

THE DECISION GAP

Today, signals live in separate worlds.

Machine alert "Something looks wrong."

Production context "What happens if it stops?"

Energy insight "Where is efficiency lost?"

WHO ACTUALLY USES THE DECISION?

Plant Manager
business consequence

Maintenance Lead
intervention window

CHALLENGE 04 • WHAT SUCCESS LOOKS LIKE

Reduce specific energy use — while protecting throughput, quality and competitiveness.

35–40%

of India's energy is consumed by industry

15–30%

of production costs can come from energy

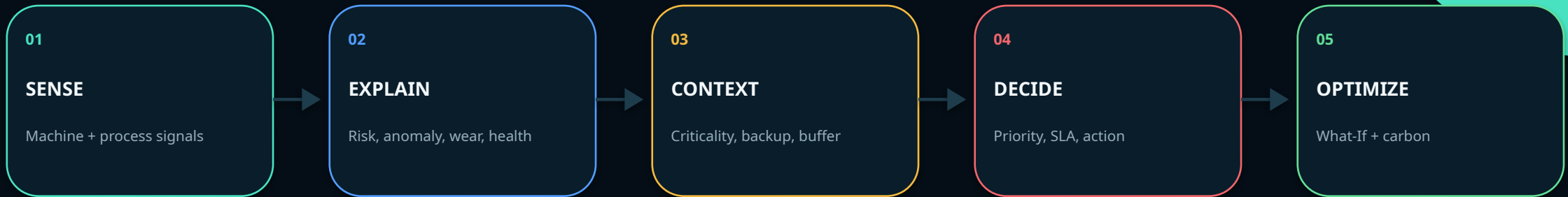
REAL-TIME

energy monitoring is missing in many SMEs

FactoryMind's positioning **Connect the four decisions in one loop.**

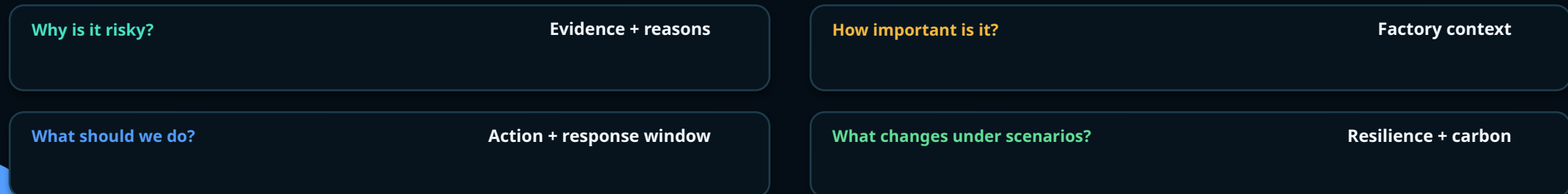
One operating loop from signal → explanation → action.

FactoryMind turns fragmented machine, operational and energy evidence into one traceable decision.



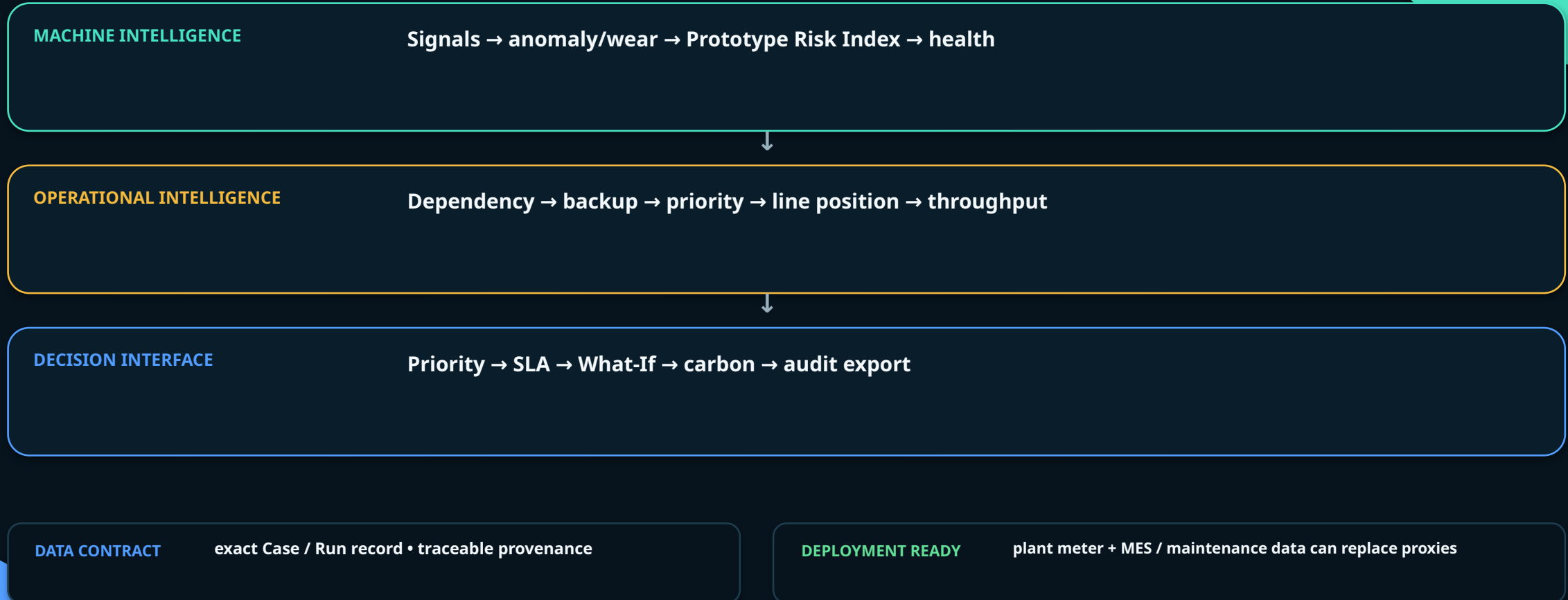
THE OUTPUT

A factory-ready answer to four questions:



A modular architecture built around “one source of truth”.

Machine intelligence is consumed as frozen evidence; the decision layer adds factory context without changing the machine state.



The operator sees the “so what?” — not the raw telemetry.

A decision-centered UX keeps the experience fast: see state → understand why → see business impact → act.



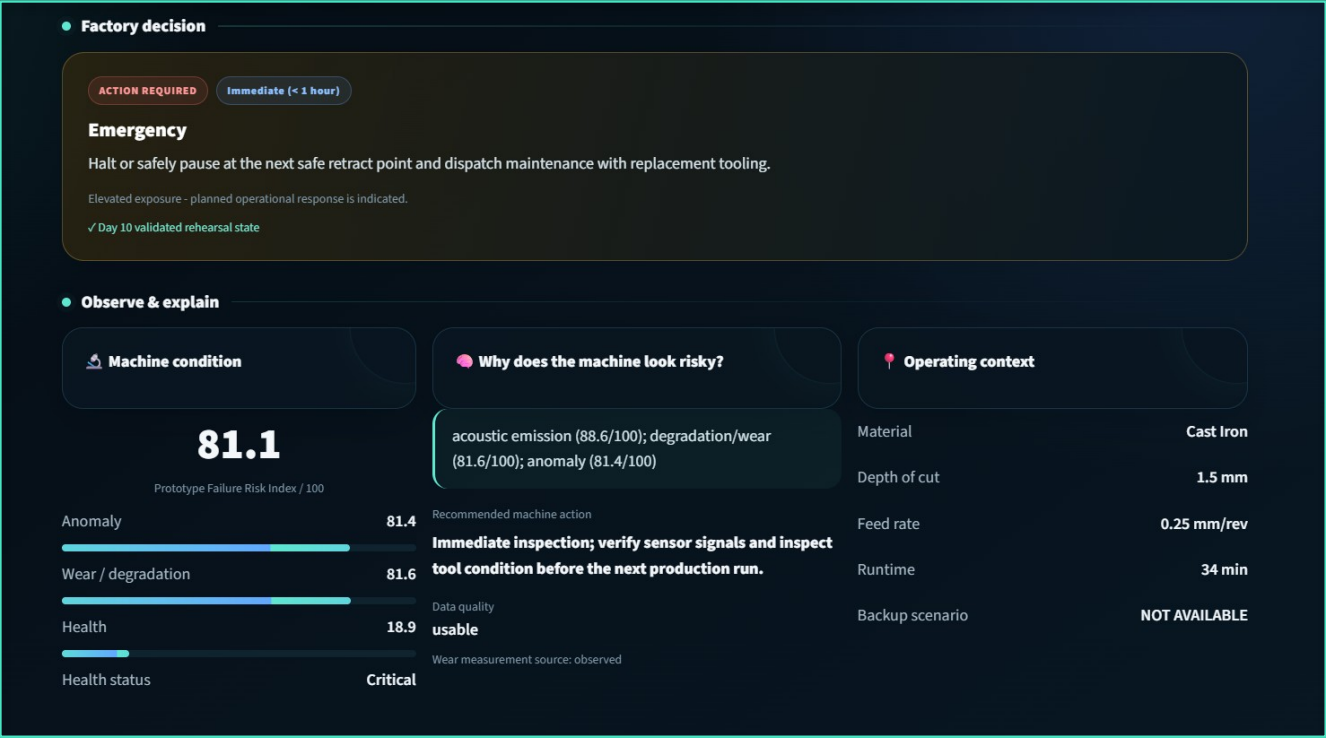
ONE SCREEN, FOUR LAYERS

Machine **condition** + Factory **consequence** + Resilience **scenario** + Carbon **lens**

No alert overload. No black box handoff. One decision moment.

The live dashboard makes the decision visible in seconds.

Wide product view: machine state, factory context, scenario controls and traceability in one command center.



DASHBOARD EVIDENCE • MACHINE CONDITION + WHY

- 01 OBSERVE
- 02 EXPLAIN
- 03 DECIDE
- 04 WHAT-IF
- 05 DECARBONIZE
- 06 AUDIT

Judge-ready mode • atomic Case/Run selector • frozen representative states

Case 4 / Run 6: strong machine evidence, then a business decision.

The machine state is frozen; operational context determines the response window.

RISK INDEX

81.1

prototype indicator

HEALTH

18.9

critical band

WHY?

Acoustic emission + wear + anomaly evidence combine into the machine-risk view.

Machine action

Verify sensor signals and inspect tool condition before the next production run.

BASELINE FACTORY SCENARIO

Emergency

Criticality 61.0 PBRI 49.5

Immediate response • < 1 hour

The machine is risky. The factory consequence makes it urgent.

ANOMALY

81.4

signal pattern

WEAR

81.6

VB ≈ 0.40 mm

Core insight: machine risk is an input to the business decision — not the decision itself.

Same machine. Different factory context. Different response.

The What-If engine tests resilience without rewriting the underlying machine evidence.

Why this matters to the factory

Operational Criticality

61.0

Tier 2: High Operational Impact

Bottleneck Exposure

49.5

MODERATE

Response

Emergency

Immediate (< 1 hour)

Deploy

> See the decision calculation

What-if resilience lab

Baseline

BackupOFF

Criticality61.0

PBRI49.5

PriorityEmergency

Active scenario

BackupOFF

Buffer30 min

Rerouting75%

ExposureCRITICAL

Line consequence: LINE STOPPAGE - downstream operations would exhaust the displayed 30-minute buffer.
Starvation risk: HIGH • The scenario changes operational context, not machine-risk prediction.

LIVE WHAT-IF VIEW

Operational consequence changes • machine evidence does not

THE BUSINESS DIFFERENCE

Same machine state • different factory response

BACKUP ON

Planned window

< 24h response

Criticality 44.8 • PBRI 36.3

BACKUP OFF

Emergency

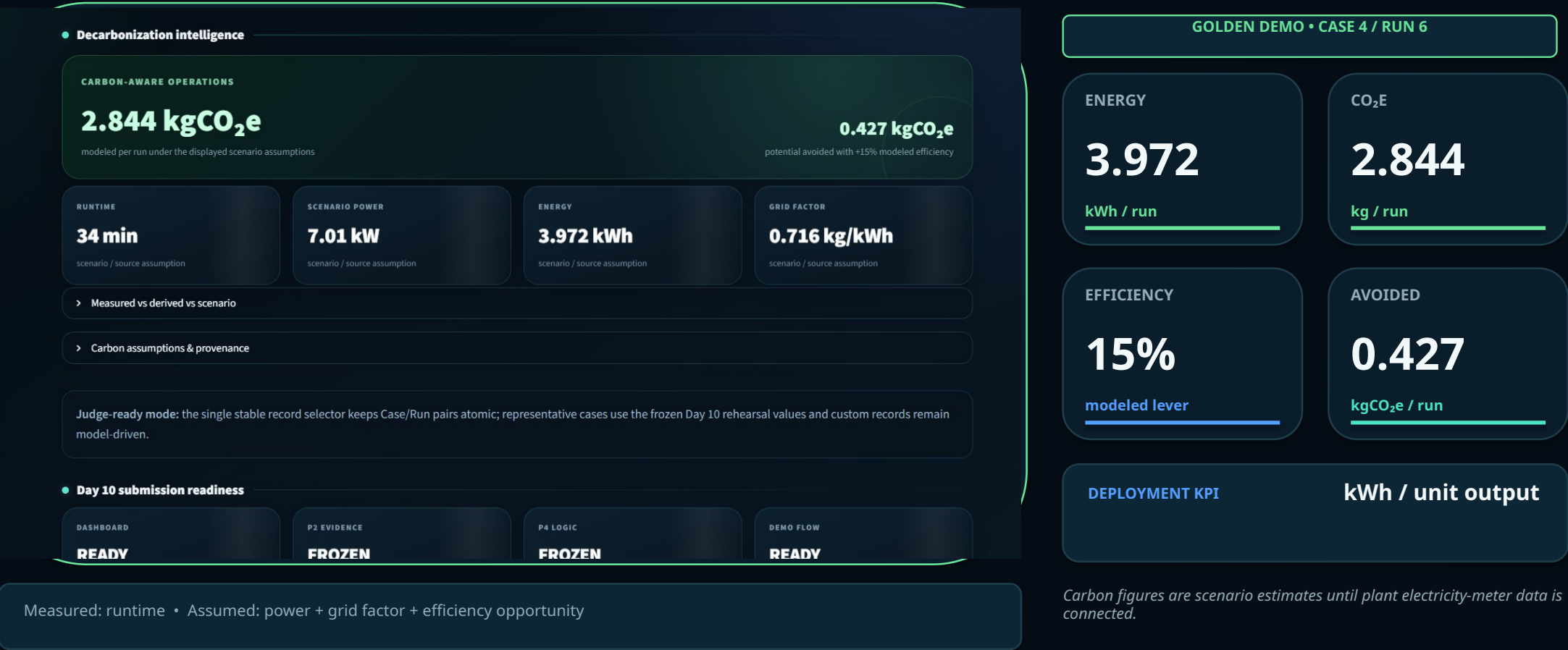
< 1h response

Criticality 61.0 • PBRI 49.5

RISK STAYS 81.1 • ONLY FACTORY CONSEQUENCE MOVES

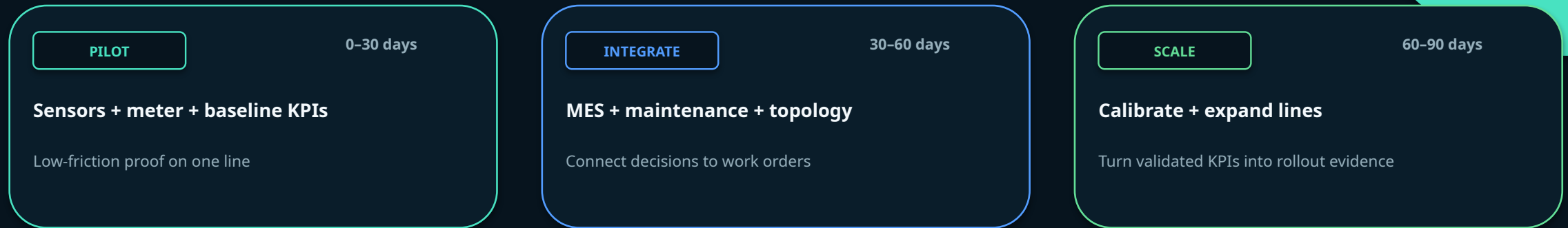
Decarbonisation is the operating outcome — not a side dashboard.

FactoryMind puts energy, maintenance and production decisions into the same workflow — with transparent scenario assumptions.



A low-friction path from decarbonisation pilot to plant-wide intelligence.

We are deliberately not investing ROI or pricing. The pilot is designed to produce the evidence needed for a real business case.



PILOT KPIs TO VERIFY ON REAL PLANT DATA



Business promise: make energy, maintenance and operational resilience measurable in the same workflow.

Built to be demonstrated, tested and trusted.

The final package is deliberately conservative about claims and explicit about provenance.

167

Case/Run records resolved

10/10

P4 stress-test checks

81.1

Golden machine state restored

61.0 →
44.8

Criticality shift with backup

VALIDATED

Atomic record selection • frozen Golden Demo • What-If invariance

TRACEABLE

Measured / derived / model / scenario labels

BOUNDED

No calibrated probability claim • no fabricated observed values

DEPLOYABLE

Scenario proxies can be replaced by plant meters + MES

JUDGE MESSAGE

We expose uncertainty, preserve machine truth, and make the decision logic auditable.

TEAM FLUXFORGE

FluxForge — four strengths, one end-to-end product.

Product, machine intelligence, operational intelligence and frontend execution were built as one system.

TEAM FLUXFORGE

RONIT GODARA

Team Lead • UI / Dashboard Design • Frontend Engineering

TEAM FLUXFORGE

YOGIK SHARMA

ML • Backend Engineering • Machine Intelligence

TEAM FLUXFORGE

RAKESH KUMAR

Operational Intelligence • Validation • Backend Integration

TEAM FLUXFORGE

SHRIKANT JOSHI

Product Strategy • Presentation • Storytelling

THE FACTORYMIND PROMISE

Predict the risk. Explain the cause. Understand the consequence. Choose the right action.

TEAM FLUXFORGE